Roll No. | | | | | | | |

EVEN END SEMESTER EXAMINATION JUNE - 2025

Name of the Course: **B.Tech** Semester: **IV**

Name of the Paper: **Mechanics of Materials** Paper Code: **TME 406**

Time: 3 Hours **Maximum Marks: 100**

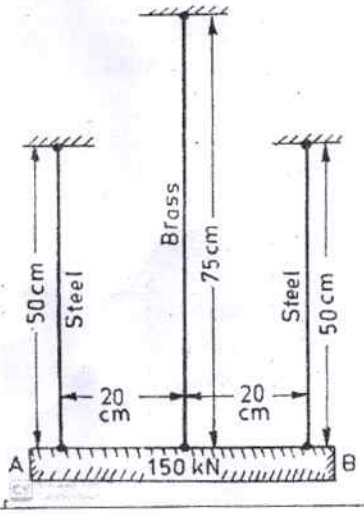
Note:

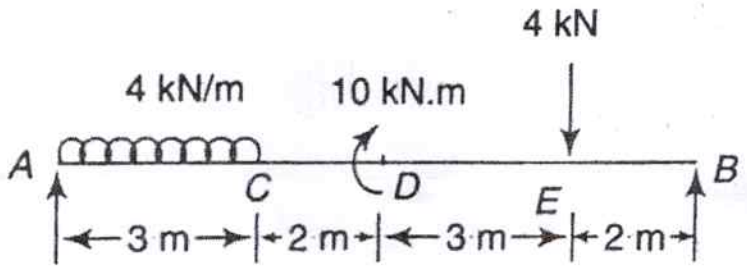
(i) All questions are compulsory

(ii) Answer any two sub questions among a, b &c in each main question

(iii) Total marks in each question are twenty

(iv) Each question carries 10 marks

Q1	**(20 marks)**	
(a)	Explain the following terms: (i) Stress (ii) Strain (iii) Poisson's Ratio (iv) Bulk Modulus (v) Modulus of Rigidity	
(b)	The stresses on two perpendicular planes through a point in a body are 30 MPa and 15 MPa both tensile along with a shear stress of 25 MPa. Find: (i) The magnitude and direction of principal stresses (ii) The planes of maximum shear stress	
(c)	A rigid bar AB is supported by three bars in the same vertical plane and are equidistant from each other as shown in figure. The outer bars are made of steel, 2.5 cm diameter and 50 cm long each. The central bar is made of brass and is 75 cm long and 2 cm diameter. Calculate the forces in the bars due to an applied force of 150 kN assuming that AB remains horizontal after the load is applied. Assume E (Steel) = 210 GPa and E (brass) = 105 GPa.  	CO1, CO2
Q2	**(20 marks)**	
(a)	A simply supported beam has distributed a load of linearly varying intensity with zero at one end to w per unit run at the other. Draw the shear force and bending moment diagrams. Also find maximum bending moment produced.	CO3

(b)	Derive the general relationship between load, shear force and bending moment in a beam.	
(c)	Draw shear force and bending moment diagram for a given beam. Also find the maximum value of bending moment.  	
Q3	**(20 marks)**	
(a)	Deduce the bending equation stating the assumptions made.	
(b)	A 120 mm wide and 10 mm thick steel plate is bent into a circular arc of 8 m radius. Determine the maximum value of stress produced. Also find the bending moment which will produce the maximum stress? E = 200 GPa.	**CO4**
(c)	Compare the weights of equal lengths of a solid and a hollow shaft to transmit a given torque for the same maximum stress if the inside diameter of the shaft is three fourth of the outside.	
Q4	**(20 marks)**	
(a)	Derive the expression for maximum slope and maximum deflection of a simply supported beam subjected to uniformly distributed load over the whole span of the beam.	
(b)	A simply supported beam has a span of 15 m and carries two-point loads of 4 kN and 9 kN at 6 m and 10 m respectively from one end. Find the deflection under each load. Take E= 200 GPa and MOI(I) = 400×10^6 mm^4.	**CO4, CO5**
(c)	What assumptions are taken in the analysis of thin cylinders? Deduce the expression for circumferential and longitudinal stresses developed in them.	
Q5	**(20 marks)**	
(a)	What is meant by crippling or buckling load? Derive the formula for Euler's crippling load for a column subjected to both ends hinged, ends conditions.	
(b)	A thin cylindrical shell 1.5 m long, 600 mm in diameter, and 8 mm thick is subjected to an internal pressure of 2.5 MPa. Calculate: (i) Hoop stress (ii) Longitudinal stress (iii) Change in diameter and length Take E = 210 GPa and Poisson's ratio = 0.28	**CO5, CO6**
(c)	A 3.2 m long fixed-end hollow cast-iron column has its internal and external diameters as 60 mm and 80 respectively. Determine Rankine's crippling load using the value of crushing stress to be 500 MPa and the value of the Rankine's constant 1/1600.	